Spectral Correlation-Based Fusion Network for Hyperspectral Image Super-Resolution

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***Abstract*— To address the limitations of hyperspectral imaging systems, super-resolution (SR) techniques that fuse low-resolution hyperspectral image (HSI) with high-resolution multispectral image (MSI) are applied. Due to the significant modal difference between HSI and MSI, and the insufficient consideration of HSI band correlation in previous works, issues arise such as spectral distortions and loss of fine texture and boundaries. In this article, an unsupervised spectral correlation-based fusion network (SCFN) is proposed to address the above challenges. A new dense spectral convolution module (DSCM) is proposed to capture the intrinsic similarity dependence between spectral bands in HSI to effectively extract spectral domain features and mitigate spectral aberrations. To preserve the rich texture details in MSI, a global–local aware block (GAB) is designed for joint global contextual information and emphasize critical regions. To address the cross-modal disparity problem, new joint losses are constructed to improve the preservation of high-frequency information during image reconstruction and effectively minimize spectral disparity for more precise and accurate image reconstruction. The experimental results on three hyperspectral remote sensing datasets demonstrate that SCFN outperforms other methods in both qualitative and quantitative comparisons. Results from the fusion of real hyperspectral and multispectral remote sensing data further confirm the applicability and effectiveness of the proposed network.**

***Index Terms*— Band correlation, dense spectral convolution module (DSCM), global–local aware block (GAB), hyperspectral image (HSI) super-resolution (SR), image fusion.**

I. Introduction

HYPERSPECTRAL imaging technology can capture dozens to hundreds of continuous spectral bands to generate hyperspectral image (HSI) with rich spectral information. Due to its powerful spectral resolution capability, it has been widely used in computer vision and remote sensing, such as object detection [1], [2], hyperspectral unmixing [3], [4], hyperspectral classification [5], [6], change detection [7], [8], and anomaly detection [9], [10]. However, the limitations of the sensor result in a tradeoff between spatial and spectral resolution in the acquired remote sensing hyperspectral data [11]. Therefore, the spatial details of remote sensing HSI may not be sufficient for some practical applications.

Super-resolution (SR) technology has a wide range of applications in the field of natural image and remote sensing image processing [12], [13], [14] and plays an important role in SR mapping [15], [16] in particular. This technology can overcome the resolution limitations of traditional imaging systems and improve the spatial resolution of images, thus making details more clearly recognizable. SR techniques for HSI can be categorized into two main groups, including single-image SR (SISR) [17], [18], [19] and multiple-image fusion SR (MIFSR) [20], [21], [23]. It depends on whether auxiliary images are used during reconstruction. Since SISR does not rely on auxiliary images with rich spatial features and considering the limited availability of training data, it becomes challenging to directly learn the mapping relationship between low-resolution (LR) images and high-resolution (HR) images [24]. MIFSR typically uses multispectral image (MSI) as auxiliary image to guide the enhancement of spatial resolution [25].

MIFSR approaches can be roughly categorized into four groups: Bayesian models [26], [27], matrix factorization-based methods [28], [29], tensor factorization-based techniques [30], [31], and deep learning approaches [32], [33]. Wei et al. [34] proposed a multiband image fusion method that utilizes Bayesian estimates to solve various fusion problems. Among the matrix decomposition-based methods, Kawakami et al. [35] applied hyperspectral unmixing algorithm to fuse RGB and HSI, and Yokoya et al. [36] introduced nonnegative features to improve the fusion problem. Furthermore, sparse representation [37], [38] and low rank feature [39], [40] have been subsequently introduced into the fusion framework for satisfactory fusion images. Matrix factorization-based methods unfold 3-D hyperspectral data into 2-D, which destroys the structure of the data. Tensor factorization offers the advantage of better preserving the data structure, which has contributed to the successful development of several influential tensor-based methods. A sparse tensor decomposition method for image fusion is introduced [41], and a nonlocal patch tensor sparse representation is designed [38]. Based on the image matting model, the regularization term is designed to preserve spectral features [42]. These model-based approaches often rely on handcrafted priors, which may introduce other difficulties when solving practical problems.

The powerful feature extraction ability of deep learning makes it have excellent performance in various image processing fields [43], [44], [45], [46], and more and more researchers pay attention to image fusion based on deep learning [47],

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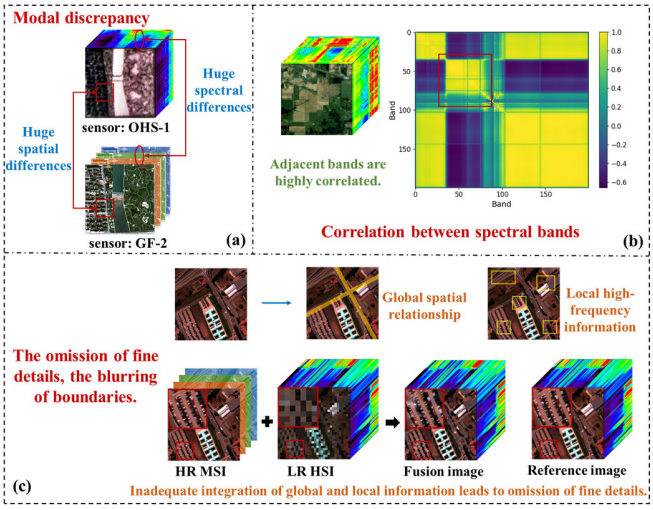


Fig. 1. Three main challenges. (a) Significant modal discrepancy leads to feature loss. (b) Underutilization of band correlation leads to spectral distortion. (c) Inadequate integration of global and local information leads to loss of fine details.

[48]. The 3-D convolutional neural networks were attempted to fuse LR HSI and HR MSI [49]. Qu et al. [50] proposed the use of networks with unsupervised encoder–decoder structures to solve the fusion problem. Iterative alternating optimization models [51] and networks with prior learning capabilities [52] have been proposed to address problems where prior knowledge is unavailable. Inspired by the wavelet decomposition-based fusion approach, Zhu et al. [53] proposed a progressive zero-centered residual network. In order to enhance the practical applications of the network, image registration and fusion have been combined and explored together [54], [55]. Taking inspiration from the unique features of a specific model, an unsupervised model-inspired HSI SR network was devised [56]. To reconcile the effects of physical models and priori information, some networks incorporates regularized optimization based on physical models [57], [58]. Li et al. [59] first proposed a new method to fuse three images, HSI, MSI, and panchromatic image (PAN). To address the problem of spectral space distortion due to improperly fabricated structures and lack of reasonable constraints, Li et al. [60] proposed an X-shaped interactive autoencoder network. Gao et al. [61] proposed an enhanced unmixing network with attentional embedding degradation learning to address the issue of insufficient constraints. A generalized fusion framework for multiple-image resolution enhancement problems was proposed by Ran et al. [62]. However, deep learning-based methods often employ complex network architectures and fail to effectively address the challenge of spatial and spectral feature loss in the presence of cross-modal discrepancies. Insufficient consideration of spectral correlations exacerbates spectral distortions.

As shown in Fig. 1, there are three main challenges in previous works for MIFSR as follows.

1) The significant modal discrepancy between HSI and MSI results in feature loss. As shown in Fig. 1(a), there are large spatial and spectral resolution differences between HSI and MSI. Li et al. [22] proposed cross-scale nonlocal attention modules to address the modal differences. However, inadequate modeling may limit the ability of the method to adapt to complex modal variations, especially in cross-sensor scenarios where modal differences are exacerbated and the complexity of the fusion problem is further revealed.
2) Underutilization of band correlation leads to spectral distortion. Fig. 1(b) highlights the similarity dependence between neighboring bands. However, general convolution operations tend to ignore this spectral correlation [56], which further exacerbates the spectral feature confusion. Although Li et al. [63] constructed spectral degradation-guided group convolution, it works independently within band groups and fails to adequately account for the overall consistency among network components.
3) Inadequate integration of global and local information leads to the omission of fine details. Concentrating solely on global spatial relationships or local high-frequency information significantly impacts the quality of fusion results [64]. As depicted in Fig. 1(c), the disparity in feature details between HSI and MSI results in blurring of the edges of certain regions in the fused image.

To solve the above problems, a spectral correlation-based fusion network (SCFN) is proposed to achieve high-quality fusion results. The network adopts an unsupervised manner and consists of two networks with encoder–decoder structure for image reconstruction of HR MSI and LR HSI, respectively. A dense spectral convolution module (DSCM) is proposed for complex spectral domain feature extraction of HSI. The global–local aware block (GAB) is designed to preserve the rich texture and edge details in MSI. New joint losses are designed to better handle cross-sensor image fusion for satisfactory image reconstruction results. Overall, the main contributions of our work are summarized as follows.

1) An SCFN is proposed for SR of hyperspectral remote sensing images. The two image reconstruction networks are interconnected by sharing decoder weights, which helps to share spectral information and combine spatial features to reconstruct quality results. Considering the problem of modal differences across sensors, reconstruction loss functions are designed for HSI and MSI, respectively, aiming at preserving the high-frequency information and minimizing the spectral differences to achieve high-fidelity image fusion results.
2) In order to enhance interband correlation utilization and mitigate spectral distortion, the DSCM is proposed aiming to efficiently extract detailed spectral features from HSI. Group convolution divides the input bands into several groups and shares weights between the bands within each group, which helps to capture the interband similarity dependence for feature enhancement and mitigation of spectral aberration. Dense connectivity allows information from all previous layers to be accessed after each grouped convolution, emphasizing the overall consistency between layers, providing greater expressive power, and helping to achieve higher quality reconstruction results.

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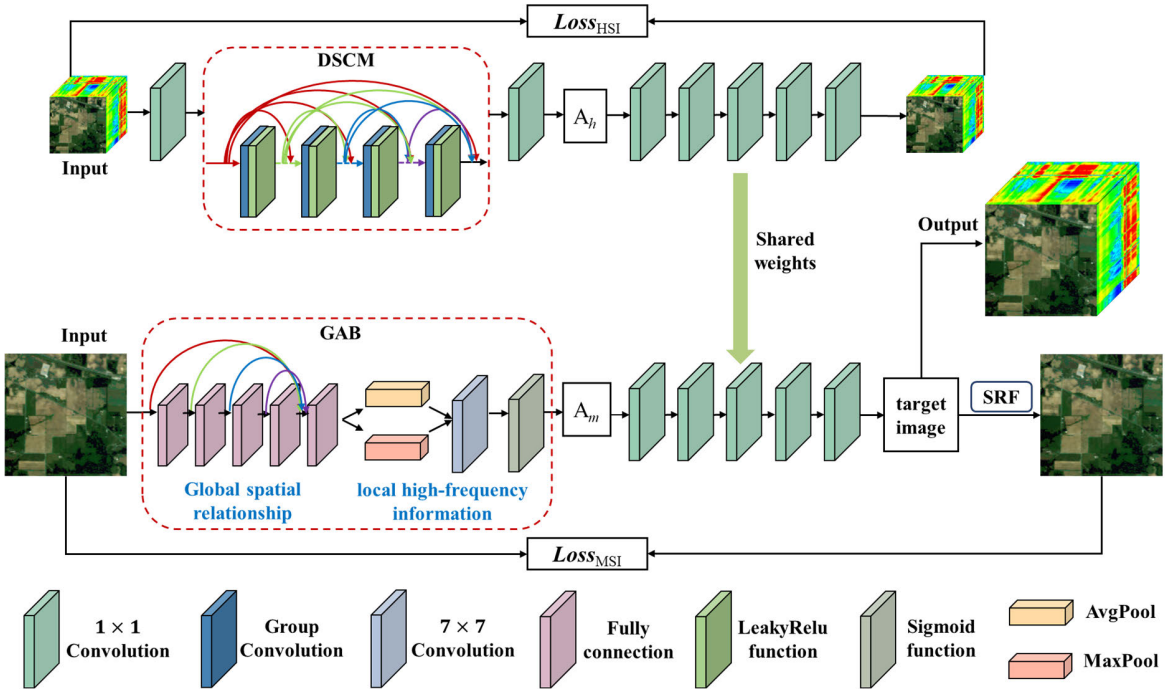


Fig. 2. Hyperspectral remote sensing image SR based on the SCFN framework.

3) To fully integrate the global spatial relations and local fine details to maintain the accuracy of complex boundaries in MSI, a simple yet effective GAB is designed as an encoder for the MSI reconstruction network. This module employs a fully connected layer with jump connections to capture global contextual information while introducing a spatial attention mechanism to emphasize fine local details in the image. This strategy aims to simultaneously satisfy the need for holistic information and local features, leading to higher quality image reconstruction.

To comprehensively evaluate the efficacy of SCFN, we conducted extensive and in-depth experiments using three publicly available datasets (Indian Pines, Salinas, and Pavia University). In addition, to verify the practical applicability of the method, fusion experiments were performed using real HSI from Orbita HyperSpectral (OHS) and MSI from Gaofen-2 (GF-2). The experimental results show that our method outperforms current state-of-the-art methods.

The rest of this article is organized as follows. Section II outlines the basic formulation of the fusion problem. In Section III, the proposed SCFN for HSI SR is described in detail. The experimental results and analysis are reported in Section IV to verify the effectiveness of our method. The conclusions are presented in Section V.

II. Problem Formulation

We denote the desired HR HSI as $\mathbf{X} \in \mathbb{R}^{M\times N\times L}$, where M, N, and L represent the length, width, and band number of HR HSI, respectively. Similarly, $\mathbf{H} \in \mathbb{R}^{m\times n\times L}$ represents LR HSI, where m and n represent the length and width of hyperspectral data, respectively. HR MSI is denoted as $\mathbf{M} \in \mathbb{R}^{M\times M\times l}$, where l represents the band number of the MSI. According to the spectral linear decomposition theory, $\mathbf{X}$ can be formulated as

$$\mathbf{X} = \mathbf{AE} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{MN\times p}$ represents abundance, $\mathbf{E} \in \mathbb{R}^{p\times L}$ represents endmembers, and p is the number of spectral bases.

Similarly, $\mathbf{H}$ and $\mathbf{M}$ can be represented separately as linear combinations

$$\mathbf{H} = \mathbf{A}_h\mathbf{E} = \mathbf{SAE} \tag{2}$$

$$\mathbf{M} = \mathbf{AE}_m = \mathbf{AER} \tag{3}$$

where $\mathbf{A}_h \in \mathbb{R}^{mn\times p}$ represents the abundance of LR HSI and $\mathbf{S} \in \mathbb{R}^{mn\times MN}$ represents PSF, which describes the spatial process. In addition, $\mathbf{E}_m \in \mathbb{R}^{p\times l}$ represents the end member of HR MSI, and $\mathbf{R} \in \mathbb{R}^{L\times l}$ represents spectral response function (SRF), which describes the spectral degradation process.

The problem of obtaining HSI with high spatial resolution through fusion processing can be described as a linear combination of $\mathbf{E} \in \mathbb{R}^{p\times L}$ in the input $\mathbf{H}$ and $\mathbf{A} \in \mathbb{R}^{MN\times p}$ in the input $\mathbf{M}$. Combined with the spectral degradation model in (3), HR MSI can be obtained from the target image, thus realizing an unsupervised process.

III. Method

The proposed fusion framework SCFN is shown in Fig. 2, which consists of two image reconstruction networks with encoder–decoder structures. Decoder weights from the HSI reconstruction network are enriched with spectral features that are shared with the MSI reconstruction network. The coding part of the MSI reconstruction network is capable of extracting rich texture structure and fine detail features, combining these features with spectral information shared

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from the decoder. This weight-sharing mechanism allows the knowledge of the rich spectral features extracted in the HSI reconstruction network to be transferred to the MSI reconstruction network. The network is able to better synthesize spectral and spatial information, resulting in more accurate and detailed reconstruction results. The proposed DSCM enhances the utilization of interband correlation, especially aiming to mitigate spectral distortion. The designed GAB can emphasize the clarity of edge details while preserving overall consistency and contextual information.

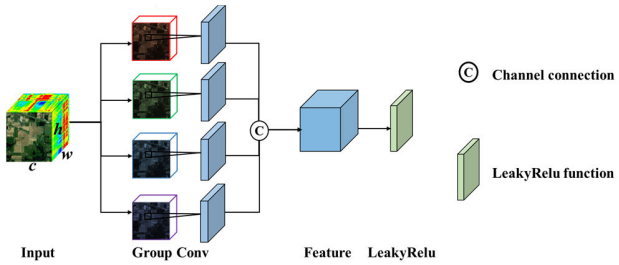


Fig. 3. Flowchart of group convolution.

A. Band Correlation Capture Guided by DSCM

The encoder of the LR HSI reconstruction network consists of two 1×1 convolution layers and the proposed DSCM. Due to the complex and redundant information contained in a large number of HSI bands and the spectral dependence between neighboring bands, the proposed DSCM incorporates the intrinsic correlation between spectral bands to extract spectral domain features and mitigate spectral distortion. The features extracted by the encoder are used to reconstruct the spectral information through a decoder, which consists of five 1×1 convolutional layers.

The proposed DSCM is composed of group convolutions, and dense connections are employed to connect the obtained feature maps along the channel dimensions. The use of group convolution not only reduces the parameter amount of the network but also prevents overfitting and achieves a similar regularization effect [65]. In addition, it can exploit the correlation between the adjacent spectral bands of HSI to avoid spectral confusion caused by normal convolution [66]. This correlation fully exploits the complementary relationships and shared characteristics among similar bands within each group. The spectral complementarity between these bands enables the mutual supplementation of information in adjacent bands, thus preserving the integrity of the overall spectral features. Combined with dense connections, the feature maps generated by each layer can be connected for reuse. The designed DSCM significantly contributes to capturing spatial–spectral features of LR HSI through intraband correlation and effectively mitigating spectral distortion.

The dimension of the feature map that goes into the regular convolution is $W \times H \times C$, where W, H, and C represent the length, width, and band number of the input image, respectively. The corresponding convolution kernel size is $k \times k \times c$, with k and c denoting the width and number of channels of the convolution kernel, respectively. However, the size of each set of feature maps input by group convolution is $W \times H \times (C/g)$, where g represents the number of groups. Similarly, the size of convolution kernel is $k \times k \times (c/g)$. Feature extraction in the form of group convolution not only reduces the amount of parameters but also fully utilizes the correlation between adjacent spectral bands to mitigate spectral distortion. In this article, the value of "g" is set to 4, as shown in Fig. 3.

Dense connection was proposed by Huang et al. [67], where each layer is capable of extracting features from all preceding layers. With dense connections, the feature maps of all previous layers are summed in the channel dimension. The subsequent convolution operation can reuse the features of the previously extracted information. This approach enhances the network's ability to extract complex spectral features. The output of this module can be expressed as

$$A_4 = \mathrm{LRelu}(G\mathrm{Conv}([A_0, A_1, A_2, A_3])) \tag{4}$$

where LRelu() represents the LeakyReLu activation function, LRelu() represents the group convolution, and [] represents the connection operation on the channel.

B. GAB for Fine Detail Preservation

The architecture of MSI reconstruction network consists of an encoder, a decoder, and a downsampling structure. The encoder is used to reconstruct the spatial features and the decoder weights are derived from the HSI branch sharing to learn the spectral information. The spectral domain of the target image is downsampled by the degradation function SRF in (3) to get the reconstructed HR MSI. The designed GAB constitutes the encoder of the MSI reconstruction network used to extract the global contextual information and edge details of the image. GAB first consists of four fully connected layers with jump connections to reduce the dimensionality of the input image while capturing the global structure. In order to generate images with clear texture details and visually pleasing, a spatial attention mechanism is introduced in this block.

Edge details play a crucial role as important structural features in images and are critical for object recognition and understanding. Based on the global context structure features obtained by the fully connected layers, maximum pooling and average pooling are conducted to generate feature maps that capture distinct information. Then, feature fusion is performed by convolution with a larger receptive field. Finally, fine local details are enhanced by overlaying the weight generated through the sigmoid operation back into the input feature map. The calculation formula of spatial attention is formulated as

$$M_s(I) = \sigma(\mathrm{Conv}([\mathrm{AvgPool}(I); \mathrm{MaxPool}(I)])) \tag{5}$$

where $M_s(I)$ represents the generated spatial attention map, σ denotes the sigmoid function, Conv() represents the 7×7 convolution operation, and $[\mathrm{AvgPool}(I); \mathrm{MaxPool}(I)]$ is feature maps generated by pooling.

The MSI decoder also consists of five 1×1 convolutions since the weights of the MSI decoder come from the shared HSI branch network. In addition, the decoder aims to restore

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features in the spectral domain, the intermediate feature representations of these two networks need to have some similarity. The specific performance is that A_h and A_m are consistent in the spectral dimension.

C. Loss Function and Training Strategy

In order to achieve more satisfactory fusion results in the cross-sensor scenarios and to employ an unsupervised training strategy, we designed loss functions for two different image reconstruction networks. Many of the previous SR reconstruction tasks used L_2 loss as the loss function, but it was not sensitive enough to small differences [33]. L_1 loss is a better choice for image reconstruction networks [68]. The network is learned and trained by minimizing the loss by calculating the loss on the input image LR HSI $\boldsymbol{H}$, HR MSI $\boldsymbol{M}$, and the two images generated by the reconstruction network.

The loss of LR HSI reconstruction network includes L_1 loss and spectral angle mapper (SAM) loss. The L_1 loss plays a significant role in guiding image reconstruction. It guides the model to reduce the absolute difference between the reconstructed and real images, facilitating the generation of high-quality results. The SAM loss further strengthens the constraints on the spectral dimension, ensuring that the reconstructed image is more consistent and accurate with the real data in terms of spectral features. The loss function composition of the HSI reconstruction network is shown by the following formula:

$$\mathcal{L}_{\text{base}}(\boldsymbol{H}, \hat{\boldsymbol{H}}) = \left\| \boldsymbol{H} - \hat{\boldsymbol{H}} \right\|_1 \tag{6}$$

$$\mathcal{L}_{\text{sam}}(\boldsymbol{H}, \hat{\boldsymbol{H}}) = \arccos\left(\frac{\boldsymbol{H} \cdot \hat{\boldsymbol{H}}}{\|\boldsymbol{H}\|_2 \left\| \hat{\boldsymbol{H}} \right\|_2} \right) \tag{7}$$

$$\mathcal{L}(\boldsymbol{H}, \hat{\boldsymbol{H}}) = \mathcal{L}_{\text{base}}(\boldsymbol{H}, \hat{\boldsymbol{H}}) + \mathcal{L}_{\text{sam}}(\boldsymbol{H}, \hat{\boldsymbol{H}}) \tag{8}$$

where $\boldsymbol{H}$ denotes the original input LR HSI and $\hat{\boldsymbol{H}}$ denotes the reconstructed LR HSI.

The HR MSI reconstruction network is mainly used to extract rich feature information in the spatial dimension and output the target image $\boldsymbol{X}$ in the network. Therefore, the reconstruction loss of the network mainly includes L_1 loss, SAM loss, and multiscale structural similarity (MS-SSIM) loss. Although brightness variations and color deviations occur, MS-SSIM loss is sensitive to high-frequency information, such as image edges and details. The L_1 loss function does a much better job of keeping luminance and color unchanged to make up for the shortcomings. The SAM loss can make the reconstructed image have less spectral distortion. The loss function composition of the MSI reconstruction network is shown by the following formula:

$$\mathcal{L}_{\text{base}}(\boldsymbol{M}, \hat{\boldsymbol{M}}) = \left\| \boldsymbol{M} - \hat{\boldsymbol{M}} \right\|_1 \tag{9}$$

$$\mathcal{L}_{\text{sam}}(\boldsymbol{M}, \hat{\boldsymbol{M}}) = \arccos\left(\frac{M \cdot \hat{\boldsymbol{M}}}{\|M\|_2 \left\| \hat{\boldsymbol{M}} \right\|_2} \right) \tag{10}$$

$$\mathcal{L}_{\text{ms-ssim}}(\boldsymbol{M}, \hat{\boldsymbol{M}}) = 1 - \left[l_m(\boldsymbol{M}, \hat{\boldsymbol{M}}) \right]^{\alpha_m} \cdot \prod_{j=1}^{m} \left[c_j(\boldsymbol{M}, \hat{\boldsymbol{M}}) \right]^{b_j} \cdot \left[s_j(\boldsymbol{M}, \hat{\boldsymbol{M}}) \right]^{g_j} \tag{11}$$



(a) (b) (c)

Fig. 4. Publicly available hyperspectral remote sensing data. (a) Indian Pines. (b) Salinas. (c) Pavia University.

$$\mathcal{L}(\boldsymbol{M}, \hat{\boldsymbol{M}}) = \mathcal{L}_{\text{base}}(\boldsymbol{M}, \hat{\boldsymbol{M}}) + \mathcal{L}_{\text{sam}}(\boldsymbol{M}, \hat{\boldsymbol{M}}) + \mathcal{L}_{\text{ms-ssim}}(\boldsymbol{M}, \hat{\boldsymbol{M}}) \tag{12}$$

where $\boldsymbol{M}$ denotes the original input HR MSI and $\hat{\boldsymbol{M}}$ denotes the reconstructed HR MSI. m indicates how many scales there are; $[l_m(\boldsymbol{M}, \hat{\boldsymbol{M}})]^{\alpha_m}$ represents the brightness calculated by the mean at the largest scale; and $[c_j(\boldsymbol{M}, \hat{\boldsymbol{M}})]^{\beta_j}$ and $[s_j(\boldsymbol{M}, \hat{\boldsymbol{M}})]^{\gamma_j}$ are the contrast calculated by the variance and the structural similarity calculated by the covariance on the jth scale, respectively.

The pseudocode of the proposed SCFN is shown in Algorithm 1.

Algorithm 1 SCFN Algorithm

Input:
LR HSI $\boldsymbol{H}$
HR MSI $\boldsymbol{M}$
Output:
Reconstructed HR HSI $\boldsymbol{X}$
1: Input H to train the LR HSI reconstruction network to obtain the decoder weights.
2: Calculate
$\mathcal{L}(\boldsymbol{H}, \hat{\boldsymbol{H}}) = \mathcal{L}_{\text{base}}(\boldsymbol{H}, \hat{\boldsymbol{H}}) + \mathcal{L}_{\text{sam}}(\boldsymbol{H}, \hat{\boldsymbol{H}})$
3: Repeat and update decoder weights.
4: Input $\boldsymbol{M}$ to train the HR MSI reconstruction network. Lock the decoder weights from the LR HSI reconstruction network and update the encoder of the HR MSI network.
5: Calculate
$\mathcal{L}(\boldsymbol{M}, \hat{\boldsymbol{M}}) = \mathcal{L}_{\text{base}}(\boldsymbol{M}, \hat{\boldsymbol{M}}) + \mathcal{L}_{\text{sam}}(\boldsymbol{M}, \hat{\boldsymbol{M}}) + \mathcal{L}_{\text{ms-ssim}}(\boldsymbol{M}, \hat{\boldsymbol{M}})$
6: Repeat and update encoder weights.
7: Until Convergence.
8: Return Target HR HSI $\boldsymbol{X}$.

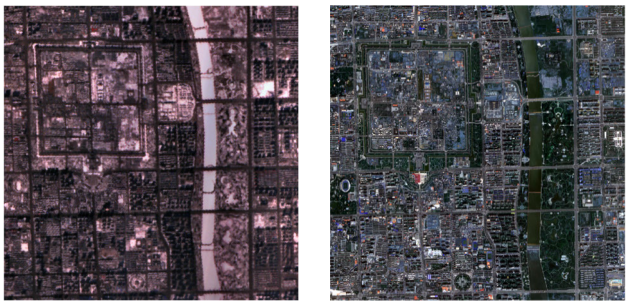
IV. Experiments

This section presents the relevant experimental analysis results based on the proposed network framework. A comprehensive evaluation of the proposed SCFN is performed on both publicly available and real scene hyperspectral data. Ablation experiments are also used to verify the advantages of the proposed modules. In addition, the analysis of urban green spaces verified the effectiveness of the proposed method.

A. Experimental Dataset

To assess the effectiveness of SCFN, experiments were conducted on three different publicly available hyperspectral remote sensing data. In addition, fusion experiments

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(a) (b)

Fig. 5. Real hyperspectral data and multispectral data. (a) OHS (bands 3, 7, and 12). (b) GF-2 (bands 1, 2, and 3).

were performed on a real hyperspectral dataset and multiple multispectral datasets. Fig. 4 shows three publicly available hyperspectral remote sensing datasets: Indian Pines, Salinas, and Pavia University [69]. The Indian Pines dataset, acquired by the airborne visible infrared imaging spectrometer (AVIRIS) sensor, spans an image area of 145 × 145 pixels. The dataset has a spatial resolution of 20 m and a wavelength range of 400–2500 nm and consists of 220 bands. In this article, a 200-band image covering a 144 × 144 pixel area is used. The Salinas dataset captured by AVIRIS sensor in CA, USA. Its spatial resolution is 3.7 m, its size is 512 × 217 pixels, and it contains 224 bands in the range of 400–2500 nm. After removing 20 bands with severe water vapor absorption, a 200 × 200 region with 204 spectral bands was cropped as the experimental image. The size of the Pavia University dataset captured by the ROSIS sensor is 610 × 340 pixels. It contains 115 bands in the 430–840-nm spectral range with a spatial resolution of 1.3 m. The affected bands were removed and an area with 103 bands of 256 × 256 pixel size was selected as the experimental data.

For real data experiments, hyperspectral data from OHS satellite and multispectral data from GF-2 are used, as shown in Fig. 5. The spatial resolutions of HSI and MSI are 10 and 4 m, and the number of bands is 32 and 4. To make the data match, the image size was slightly adjusted to 428 × 428 for HSI and 1284 × 1284 for MSI, with a fusion ratio of 3.

B. Quantitative Evaluation Metrics

For publicly available datasets, quantitative performance is used to compute and evaluate each algorithm. Specifically, peak signal-to-noise ratio (PSNR) [70] for spatial error evaluation and spectral angle mapping (SAM) for spectral comparison are included. PSNR is a commonly used error evaluation index in the field of image SR reconstruction. Usually, we generally believe that higher PSNR values mean better image reconstruction results. The PSNR value is calculated from the mean square error (MSE) between the ground truth and the estimated value. The MSE calculation formula is as follows:

$$\mathrm{MSE} = \frac{1}{mnl}\sum_{i,j,k}(I - \hat{I})^2 \tag{13}$$

where m, n, and l, respectively, represent the length, width, and number of channels of the image, respectively; I represent the real ground HR HSI; and $\hat{I}$ represents the SR HSI generated by the network. According to the calculated MSE, root MSE (RMSE) and PSNR can be expressed by the formula

$$\mathrm{RMSE} = \sqrt{\mathrm{MSE}} \tag{14}$$

$$\mathrm{PSNR} = 10 \cdot \log_{10}\left(\frac{\mathrm{Max}(I)^2}{\mathrm{MSE}(I, \hat{I})}\right). \tag{15}$$

SSIM is a metric used to compare the degree of similarity between two images, taking into account a combination of information such as brightness, contrast, and structure. When the calculated SSIM value is closer to 1, the higher the similarity between the reconstructed image and the original image, and the better the reconstruction effect. The formula of SSIM is as follows:

$$\mathrm{SSIM} = \frac{(2\mu_I\mu_{\hat{I}} + c_1)(2\sigma_{I\hat{I}} + c_2)}{(\mu_I^2 + \mu_{\hat{I}}^2 + c_1)(\sigma_I^2 + \sigma_{\hat{I}}^2 + c_2)} \tag{16}$$

where μ_I and $\mu_{\hat{I}}$ are the pixel means of the reference and reconstructed images, respectively; μ_I^2 and $\mu_{\hat{I}}^2$ denote the pixel variance of the two images, respectively; $\sigma_{I\hat{I}}$ is the pixel covariance of the two images; and c_1 and c_2 are constants to prevent the denominator from being 0.

SAM is an important measure of the spectral difference between two images and is used to calculate the angular difference between the spectral vectors between the two. In general, the lower the SAM value, the smaller the spectral angle difference between the two images, and the less spectral distortion. The calculation process of SAM is shown as follows:

$$\mathrm{SAM} = \arccos\left(\frac{I \cdot \hat{I}}{\|I\|_2\|\hat{I}\|_2}\right). \tag{17}$$

The SAM of the entire image is the average of the SAMs of all pixels in the image.

C. Implementation Details

For the simulation experiments, the LR HSI is obtained by constructing a Gaussian filter with the same width as the ground sample distance (GSD) ratio and a value deviation of 0.5 for the original HR HSI. Downsampling in the spectral domain using SRF as a filter was employed to obtain the HR MSI. The SRF is estimated using the spectral response from the IKONOS satellite corresponding to each band of the MSI as a reference and processed using B-spline interpolation to obtain the same number of bands as the input hyperspectral data. The GSD ratio of the three simulation datasets was set to 8, and all images were normalized between 0 and 1. We implemented the proposed network framework based on the TensorFlow framework and trained it on four NVIDIA RTX2080 graphics processing units (GPUs). The Adam optimizer is used to train the proposed network by initializing the learning rate to 0.001. The proposed SCFN is compared with eight other advanced fusion methods, including smoothing filter-based intensity modulation (SFIM) [71], Gram–Schmidt adaptive (GSA) [72], coupled nonnegative matrix factorization (CNMF) [36], HySure [73], fast fusion based on Sylvester equation (FUSE) [34], generalized laplacian pyramid hyperspectral (GLPHS) [74], coupled sparse tensor factorization

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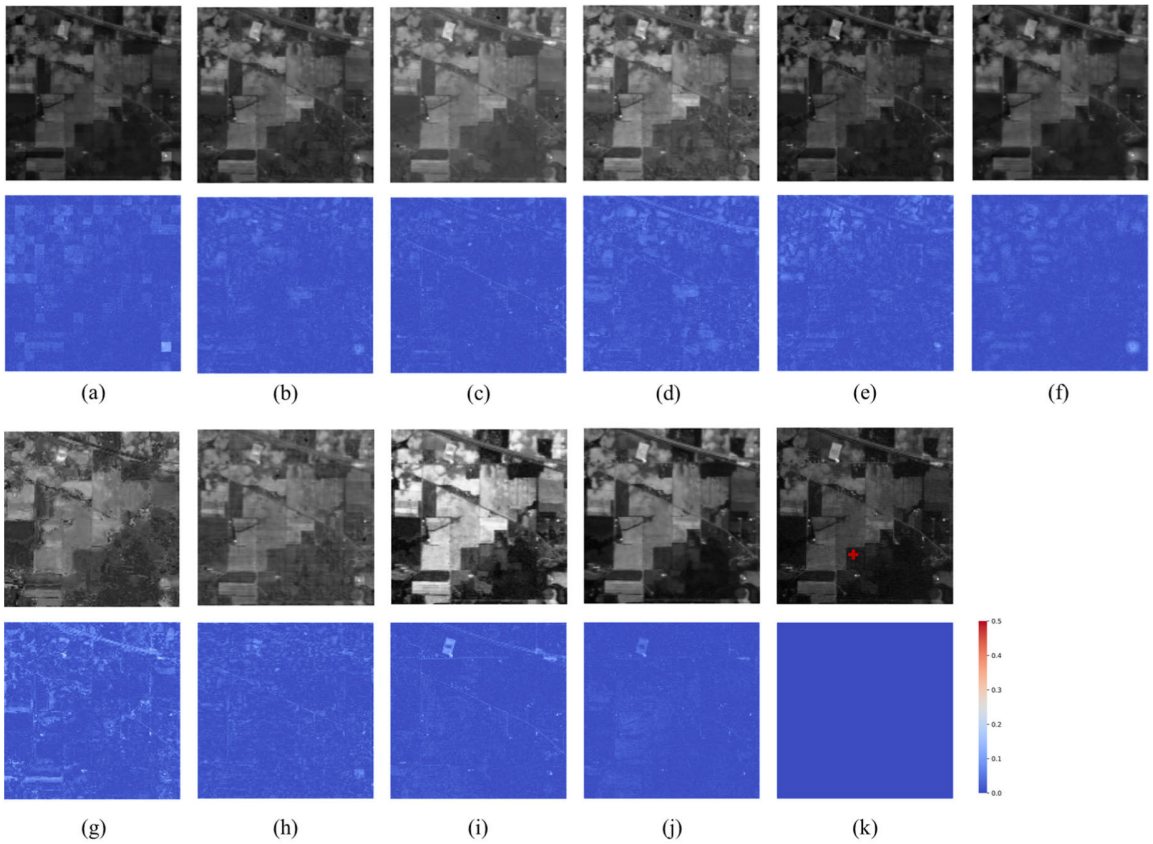


Fig. 6. Grayscale image and error map of each method at band 5 on the Indian Pines dataset. Methods (a) SFIM, (b) GSA, (c) CNMF, (d) HySure, (e) FUSE, (f) GLPHS, (g) MIAE, (h) CSTF, (i) uSDN, (j) SCFN, and (k) reference image.

TABLE I

QUALITY EVALUATION OF INDIAN PINES DATASET, WITH THE BEST RESULTS SHOWN IN BOLD

	SFIM	GSA	CNMF	HySure	FUSE	GLPHS	MIAE	CSTF	uSDN	SCFN
PSNR	30.252	32.715	32.760	34.180	30.226	31.931	34.642	35.210	35.668	**38.037**
RMSE	0.023	0.021	0.017	0.021	0.026	0.019	0.062	0.017	0.016	**0.012**
SSIM	0.932	0.945	0.946	0.916	0.929	0.947	0.799	0.944	0.946	**0.965**
SAM	3.042	2.748	2.487	2.889	3.221	2.699	9.678	2.778	2.388	**1.603**

(CSTF) [41], unsupervised sparse dirichlet-net (uSDN) [50], and model inspired autoencoder (MIAE) [56]. For methods that cannot adaptively learn the SRF but need to use it for optimization, such as uSDN, the estimated SRF is used as a known prior.

D. Results on Indian Pines

Fig. 6 shows the qualitative visualization results and error plots for each method on this data. The fusion outcomes of SFIM, GSA, and GLPHS show obvious patchy phenomena. The CNMF, MIAE, and FUSE methods do not perform well in distinguishing the boundaries of different classes of features. HySure and CSTF methods introduce additional noise. The uSDN resulted in overall higher pixel values and large differences in spectral eigenvalues from the reference image. The proposed SCFN obtains fusion results closest to the reference image while exhibiting clear textures and distinct differentiation of feature boundaries.

Since some less significant nuances are not easily perceived by humans, we also compute error maps between the estimated results and the reference image when visualizing the fusion results. These error plots in Fig. 6 give a clearer picture of how the performance of each method compares. There are obvious block errors in the SFIM and GLPHS methods. For the feature with irregular texture, the GSA, FUSE, and HySure have weak perceptual ability. The CNMF, MIAE, CSTF, and uSDN cannot accurately capture the feature boundary information. The GAB designed in the proposed method can effectively combine global context information and boundary features to obtain better fusion results.

To assess the ability of each method in maintaining the spectral characteristics, we plot spectral curves at the red marker for each result, as shown in Fig. 7. The results show that the spectral profile obtained by the proposed SCFN is the closest to the ground truth, demonstrating its advantage in spectral reconstruction. In addition to qualitatively comparing the results obtained by each method, we also performed a quantitative evaluation of the data results. Table I presents the quantitative evaluation of all comparative methods on the Indian Pines dataset. In the case that the evaluation indicators obtained by most traditional methods are not ideal, the proposed SCFN still achieved the best performance on all

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TABLE II
QUALITY EVALUATION OF SALINAS DATASET, WITH THE BEST RESULTS SHOWN IN BOLD

	SFIM	GSA	CNMF	HySure	FUSE	GLPHS	MIAE	CSTF	uSDN	SCFN
PSNR	32.910	40.148	43.124	35.024	36.469	37.721	36.136	40.218	45.673	**48.148**
RMSE	0.023	0.010	0.007	0.018	0.015	0.013	0.063	0.010	0.005	**0.004**
SSIM	0.960	0.990	0.994	0.955	0.980	0.986	0.995	0.988	0.995	**0.998**
SAM	2.365	2.131	1.209	4.002	2.431	1.915	16.737	2.042	2.620	**0.916**

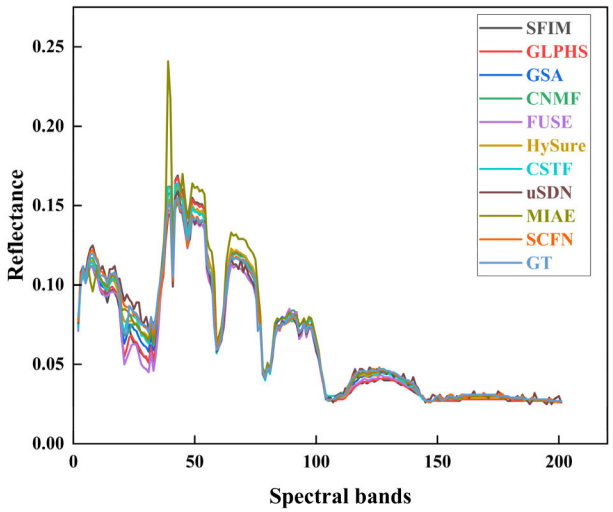


Fig. 7. Spectral profile of each result at the red marker for Indian Pines.

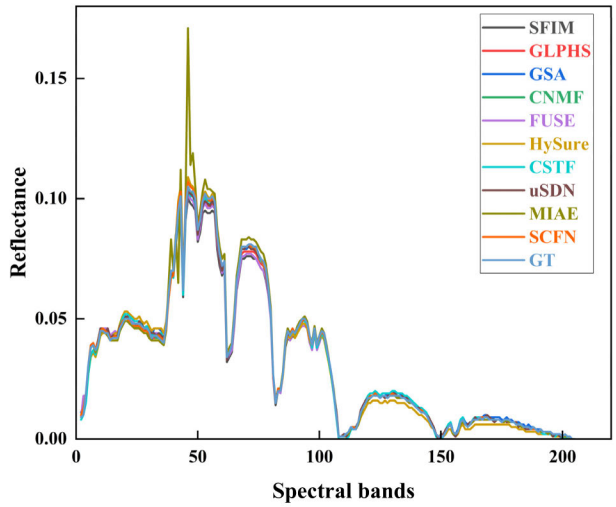


Fig. 8. Spectral profile of each result at the red marker for Salinas.

evaluation metrics, with a PSNR value of 38.037, an RMSE value of 0.012, an SSIM value of 0.965 closest to 1, and an SAM value of the smallest 1.603. Although some methods achieve high PSNR values, their SAM values are much higher than ours and do not restore a satisfactory spectral structure. In contrast, the DSCM designed in our method is able to fully utilize the relationship between adjacent bands to capture the intrinsic correlation of spectra and effectively mitigate the spectral aberration phenomenon.

E. Results on Salinas

The fusion results of SFIM, FUSE, GLPHS, and CSTF exhibit chunking, resulting in insufficient image boundaries and image spots inside the image and unsatisfactory quality of fusion SR reconstruction. These algorithms lack effective modeling of the local structure of the image, rely too much on global spatial relationship modeling, and are unable to fully capture and utilize the complex texture features of the image boundary regions as well as the internal detail regions. The results generated by SFIM, FUSE, HySure, and MIAE have a large difference between the gray value at the 51 band and the real image, and there are obvious color differences. GSA, CNMF, uSDN, and the method proposed SCFN have better boundary feature expression and satisfactory fusion results.

The reconstruction errors of SFIM, GSA, FUSE, GLPHS, MIAE, and CSTF are relatively large, although we think that the results of GSA are excellent in the previous grayscale display. The recovery ability of each method in detail can be further judged from the error graph. For example, there are still some imperceptible noises in the CNMF reconstruction results. uSDN is less aware of boundary features than the proposed method. On the whole, the GAB designed in this network has advantages in recovering image texture structure and detail features. The spectral curves at the red marks in each result graph are shown in Fig. 8, and the result of the proposed method is closer to the real HR HSI. Fig. 9 shows the grayscale display pictures and corresponding error plots for each method in band 51 of the Salinas data.

Table II presents the comparative results, with the proposed method outperforming others with PSNR, RMSE, SSIM, and SAM values of 48.148, 0.004, 0.998, and 0.916, respectively. It is demonstrated to be superior in preserving spatial structure and spectral information. Upon closer inspection, it is easy to see that uSDN has the worst SAM value despite its higher PSNR value. This indicates that the method has a high ability to capture spatial structure, but complex spectral features are difficult to recover. The DSCM in the proposed method contributes significantly to the excellent fusion results by effectively capturing the intrinsic relationship between the bands and mitigating the spectral aberrations.

F. Results on Pavia University

It presents a large number of complex boundary and detail features with more varied and complex textures. The results of SFIM still have serious block effects; GLPHS, CNMF, and CSTF show inaccurate representation of surface objects; and CTSF produces a lot of noise. The gray values obtained by FUSE, CSTF, and uSDN do not match the ground truth. The

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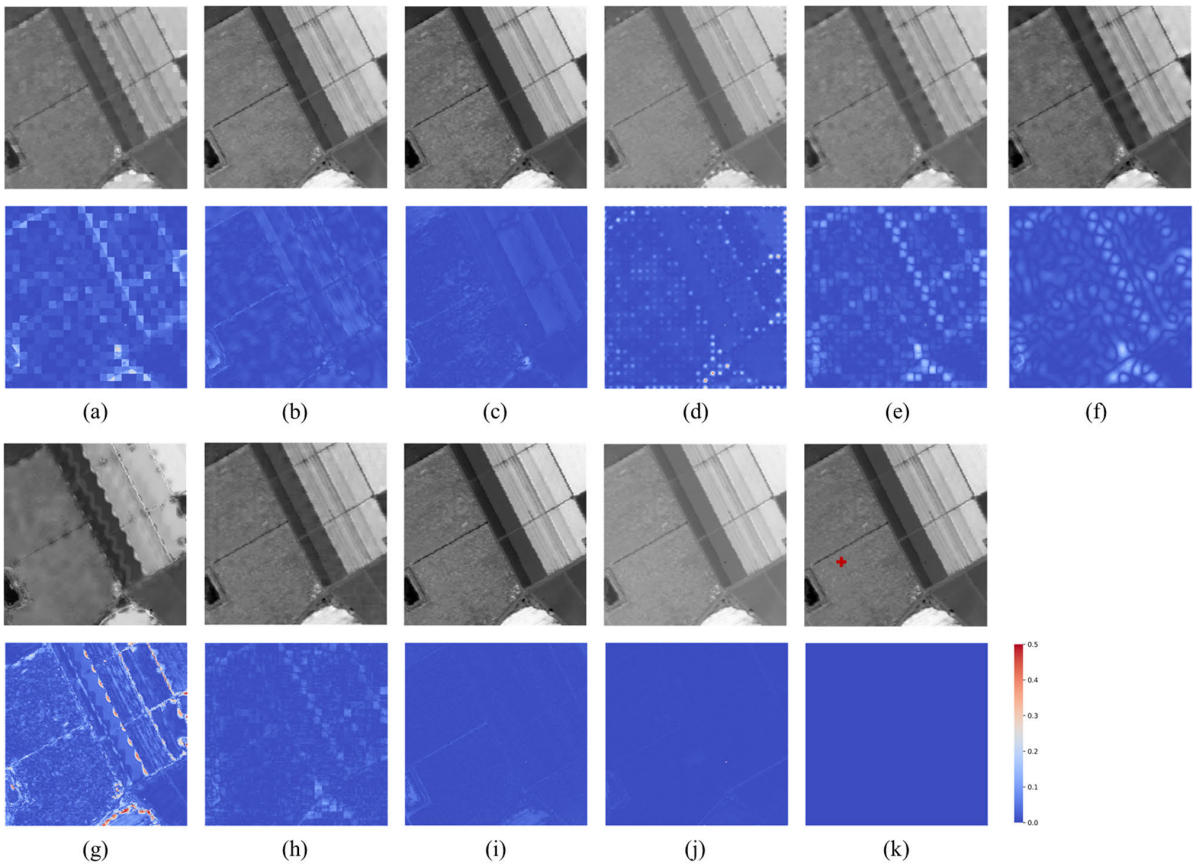


Fig. 9. Grayscale image and error map of each method at band 51 on the Salinas dataset. Methods (a) SFIM, (b) GSA, (c) CNMF, (d) HySure, (e) FUSE, (f) GLPHS, (g) MIAE, (h) CSTF, (i) uSDN, (j) SCFN, and (k) reference image.

TABLE III
QUALITY EVALUATION OF PAVIA UNIVERSITY DATASET, WITH THE BEST RESULTS SHOWN IN BOLD

	SFIM	GSA	CNMF	HySure	FUSE	GLPHS	MIAE	CSTF	uSDN	SCFN
PSNR	23.714	28.091	29.112	33.278	24.541	27.849	**42.575**	30.246	33.347	39.622
RMSE	0.076	0.040	0.035	0.022	0.060	0.041	**0.007**	0.033	0.021	0.011
SSIM	0.843	0.941	0.951	0.975	0.880	0.933	**0.996**	0.945	0.979	0.994
SAM	8.759	10.178	6.333	4.189	12.941	7.011	**2.175**	7.408	6.342	2.683

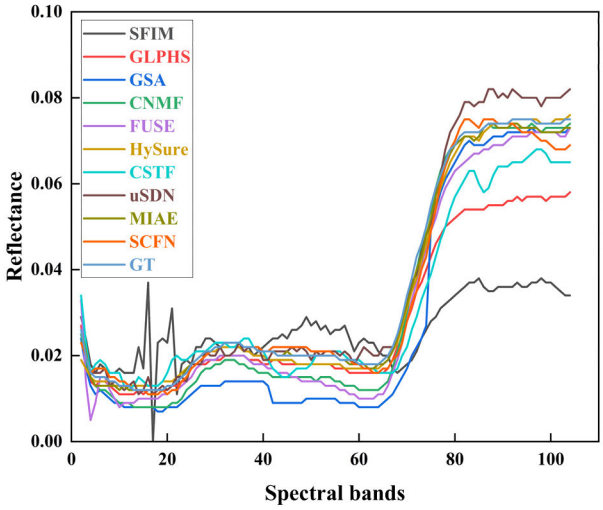


Fig. 10. Spectral profile of each result at the red marker for Pavia University.

results of GSA have problems such as blurred boundaries and incomplete expression of details. HySure, MIAE, and the proposed method achieve excellent performance on visual effects.

There are large errors between the results of the model-based approach and the reference image, and the block effect can be clearly observed. The overall error of the results obtained by HySure and uSDN still has some errors in some areas with high pixel values. There is almost no error between the results obtained by MIAE and the real image. The results obtained from the proposed SCFN have almost no error with the real image, indicating the advantages of the method in extracting and preserving the spatial texture feature representation from MSI. The spectral curves shown in Fig. 10 point out that the proposed method is effective in the retention of spectral features. Pavia University was used to further validate the performance of the proposed framework, and the grayscale display and error maps of the 31th band are shown in Fig. 11.

Table III shows the quantitative comparison results of various methods on Pavia University data. The proposed SCFN obtained the PSNR value of 39.622, the RMSE value of 0.011, the SSIM value of 0.994, and the SAM value of 2.683.

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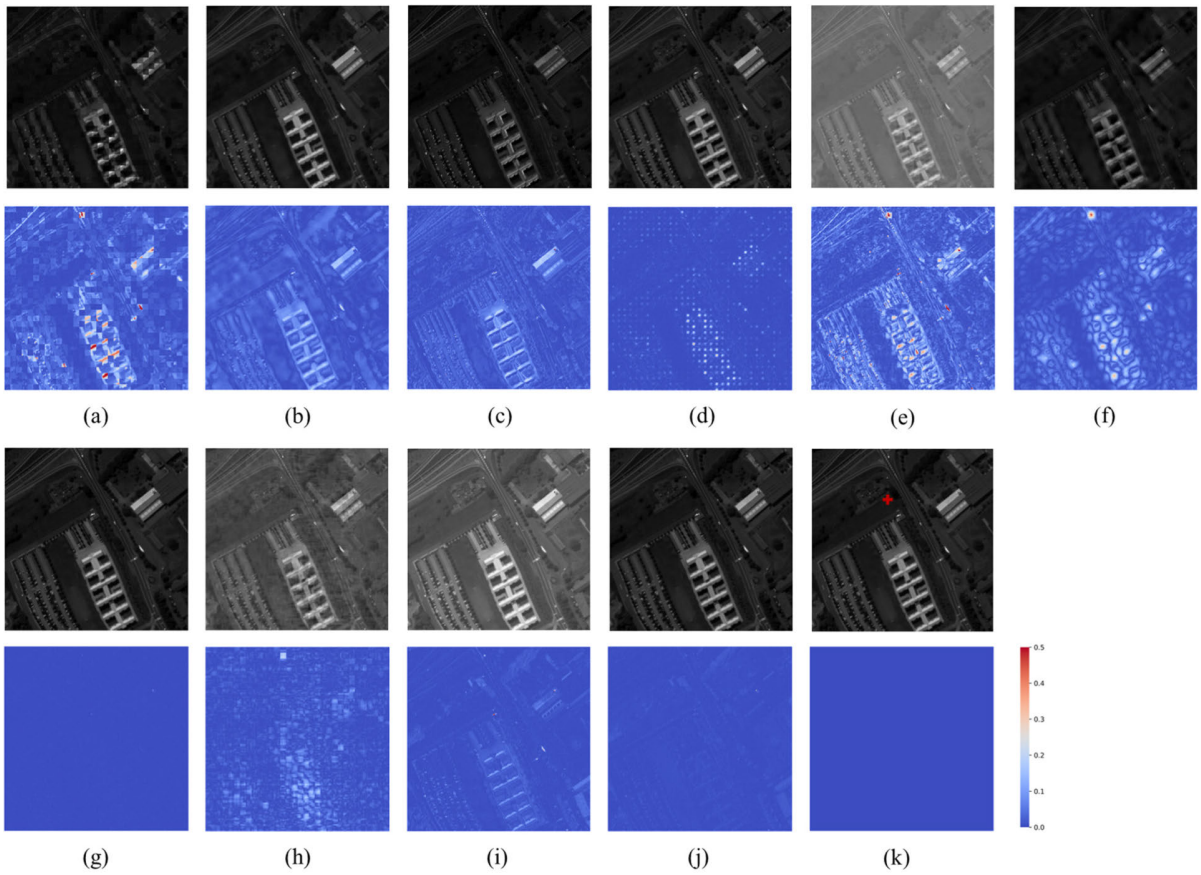


Fig. 11. Grayscale image and error map of each method at band 31 on the Pavia University dataset. Methods (a) SFIM, (b) GSA, (c) CNMF, (d) HySure, (e) FUSE, (f) GLPHS, (g) MIAE, (h) CSTF, (i) uSDN, (j) SCFN, and (k) reference image.

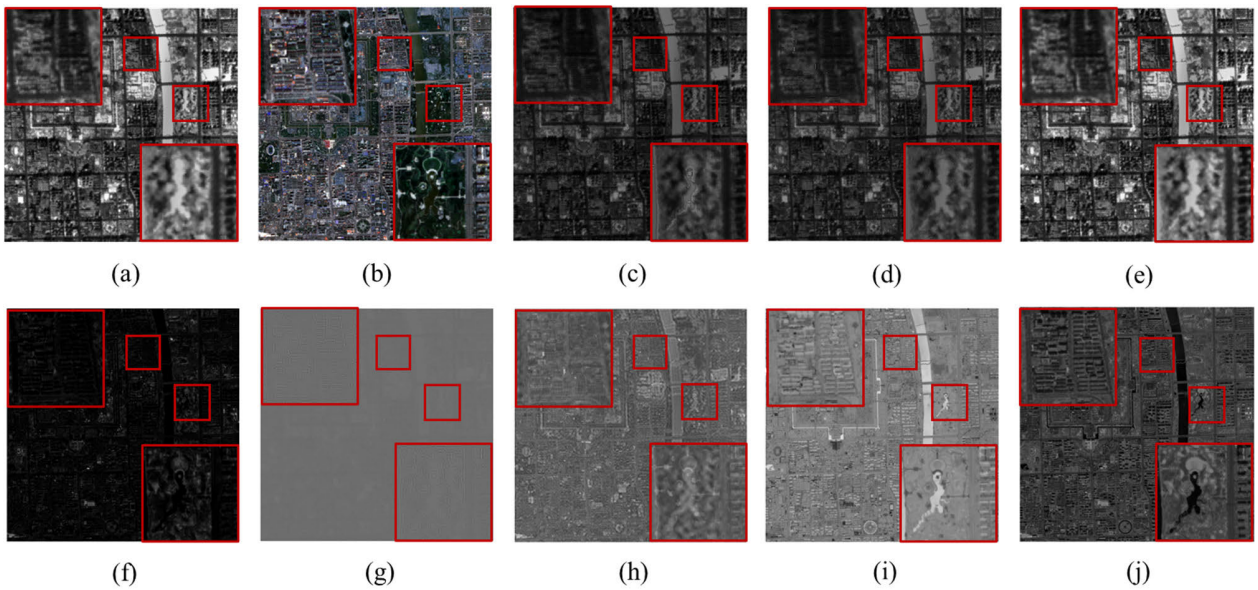
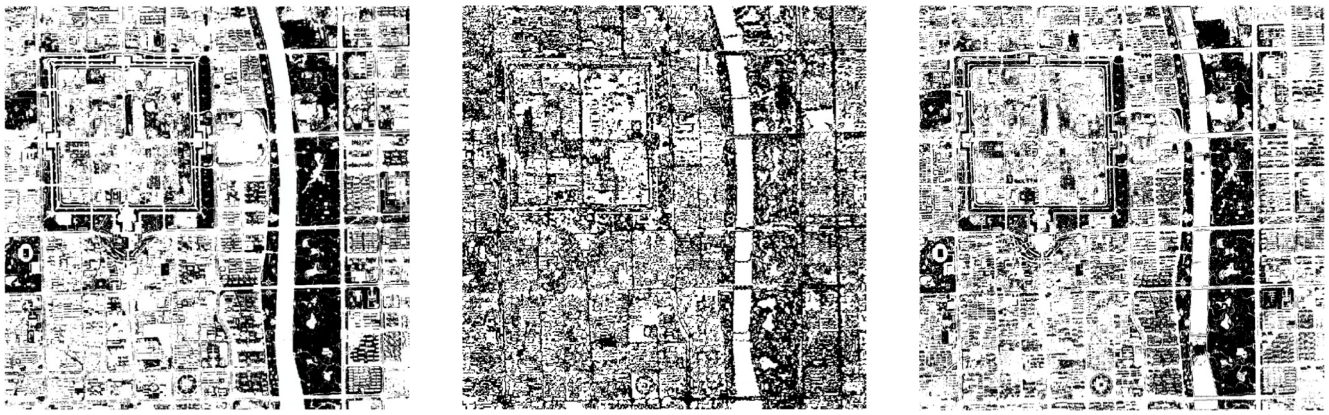


Fig. 12. Visualization of the 14th band of reconstruction results on real data. Methods (a) LR HSI, (b) HR MSI (bands 1, 2, and 3), (c) SFIM, (d) GSA, (e) CNMF, (f) HySure, (g) FUSE, (h) GLPHS, (i) uSDN, and (j) SCFN.

Furthermore, the reconstructed images exhibit clear textures and more accurate spectral structures. Except for HySure, the PSNR and SAM values obtained by the remaining pan sharpening methods, matrix decomposition-based methods, and tensor-based methods are not satisfactory. The PSNR and SSIM values calculated by the deep learning method uSDN are appropriate, but the high SAM value indicates that the recovery ability in the spectral domain is insufficient. Although MIAE

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(a) (b) (c)

Fig. 13. Results of the analysis of urban green space coverage on the real dataset. (a) HR MSI. (b) LR HSI. (c) SCFN.

TABLE IV
ABLATION EXPERIMENT RESULTS OF THE PROPOSED METHOD ON THE PAVIA UNIVERSITY DATASET

Method	DSCM	GAB	Joint loss	PSNR	RMSE	SSIM	SAM
(a)SCFN w/o DSCM	-	√	√	34.086	0.020	0.977	4.050
(b)SCFN w/o GAB	√	-	√	36.499	0.0155	0.988	3.477
(c)SCFN w/o loss	√	√	-	38.014	0.012	0.991	3.082
(d)SCFN	√	√	√	**39.622**	**0.011**	**0.994**	**2.683**

achieves optimal performance on this dataset, the first two sets of experiments allow us to learn that its performance is limited when dealing with a higher number of bands. On this dataset, most of the methods have high SAM values, and the spectral features are very different from the real data, with large spectral aberrations. The proposed method achieves a significantly small spectral angle error, thanks to the designed DSCM, which effectively leverages interband correlations and mitigates spectral distortions.

G. Experiments With Real Datasets

To verify the applicability of SCFN in real-world scenarios, real data from OHS and GF-2 satellites were used for the experiment. These experimental results will provide strong support and validation of the feasibility and practicality of our method. Due to the lack of reference images, only qualitative visualization images were taken to evaluate the results, as shown in Fig. 12. The SFIM and HySure results produced artifacts, GSA, CNMF, and GLPHS methods were not effective in improving resolution, and FUSE lost a large number of object features. As a traditional fusion algorithm based on statistical models and optimization strategies, FUSE is difficult to adequately capture the rich metric spatial–spectral features in large-size complex images. Both uSDN and SCFN provide tremendous spatial resolution improvements to the images, while SCFN exhibits sharper edges and richer detail features, with potential applications in real remote sensing scenarios with complexity.

H. Analysis of Urban Green Space Coverage

To deeply validate the capability of the proposed SCFN for practical applications in remote sensing, we conducted an experiment to analyze the urban green space coverage. The proposed method was applied to the real dataset and the generated fusion results were computed. Fig. 13(a)–(c) represents the results of the green space coverage calculated using HR MSI, LR HSI, and SR HSI, respectively. Compared with the original LR HSI, the reconstructed image generated by SCFN successfully provides clearer information about the green space coverage. In the experimental area, the distribution of green areas presents certain spatial characteristics. Specifically, the vegetation is mainly concentrated on the right side of the river, while the green space coverage in the urban area is limited. Considering the general pattern of green space distribution in the city, the overall green coverage is not significant.

I. Ablation Experiments

Ablation experiments are performed to verify the efficacy of each component in the proposed framework, which includes DSCM, GAB, and joint loss function. Pavia University is chosen for the ablation experiments because it has more complex scenarios and boundary details. Table IV shows the detailed results of the ablation experiments. In the ablation experiments, we replaced the components of the proposed framework one by one. Table IV(a) shows the results after replacing the DSCM with four convolutional layers, Table IV(b) indicates that the GAB is replaced by four fully connected layers, and Table IV(c) uses L_1 loss to replace the joint loss function. The proposed DSCM is important for capturing the correlation between spectral bands, effectively extracting spectral domain features, and mitigating spectral distortion. The designed GAB is proficient in retaining the global contextual information of the image and emphasizes the ability to

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perceive the edge details. The joint loss not only provides an effective training strategy for unsupervised learning but also proves beneficial for solving cross-sensor image fusion problems. When these components are coupled together, the performance of fused SR reconstruction is significantly enhanced.

V. Conclusion

In this article, a new unsupervised SCFN is proposed to fuse HSI and MSI. The proposed SCFN consists of two networks with an encoder–decoder structure for image reconstruction of LR HSI and HR MSI. The two image reconstruction networks are coupled together in a way of sharing decoder weights to extract and recover the spectral features of HSI and the spatial structure of MSI. The proposed DSCM combines the intrinsic links between bands to fully utilize the interband correlation. It enhances the ability of extracting spectral domain features and avoids spectral distortion as much as possible. The designed GAB effectively preserves the global contextual information of the image and emphasizes the ability of perceiving edges in order to obtain a satisfactory texture structure with clear boundaries. Joint loss functions are proposed for the two image reconstruction networks separately to train the networks to generate SR HSI with pleasing visual appearance and convincing quantitative assessment results. Experimental results on hyperspectral remote sensing data from three public and one real scene demonstrate the efficacy and adaptability of SCFN. Compared to other state-of-the-art methods, the proposed method consistently achieves realistic textures and sharp edge details with excellent spectral domain recovery performance. However, the current method needs to be trained for new fused data each time when applied, which limits the practical application scope and efficiency of the method. It is recommended to explore the construction of more universal training models to improve the adaptability and scalability of the method.

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